CLOUD-BASED WEATHER MONITORING & FORECAST DASHBOARD

Project Documentation | Using Python and Streamlit

1. Introduction

The Cloud-Based Weather Monitoring and Forecast Dashboard is a web application developed using Python and Streamlit. It retrieves weather information from the OpenWeather API and presents current conditions and a five-day forecast in an easy-to-understand interface. The application also supports visual analysis, forecast downloads and cloud-based storage of weather history through Firebase Firestore.

2. Problem Statement

Users need a convenient way to view current weather conditions and upcoming forecasts without manually collecting information from multiple sources. They may also need to compare weather patterns, download forecast data and retain previous observations. This project brings these activities together in one dashboard.

3. Objectives

- Retrieve current weather details for a user-selected city.
- Display forecast information for the next five days.
- Present weather data using tables, metrics and interactive charts.
- Allow users to export forecast data in Excel and CSV formats.
- Save and retrieve weather history using Firebase Firestore.

4. Technologies Used

Technology	Purpose
Python and Streamlit	Application logic and interactive web interface
OpenWeather API and Requests	Fetching current weather and forecast data
Pandas	Cleaning, processing and organising data
Plotly	Interactive charts and visualisation
Firebase Admin SDK / Firestore	Cloud storage and retrieval of weather history
OpenPyXL	Generating Excel forecast downloads

5. Working Methodology

The user enters a city name in the Streamlit interface. Python sends a request to the OpenWeather API and receives weather data in JSON format. Pandas processes the forecast records, and Streamlit displays the results as weather metrics, tables and charts. The user can download the forecast as a CSV or Excel file. When requested, the application uses the Firebase Admin SDK to save weather details in a Firestore collection and retrieve previous records.

6. Results and Conclusion

The application brings weather monitoring, forecast visualisation, data export and cloud history into one interface. It demonstrates the practical use of API integration, Python data processing, interactive dashboard development and cloud database connectivity. Future improvements may include user authentication, weather alerts, map-based views and more advanced historical trend analysis.

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